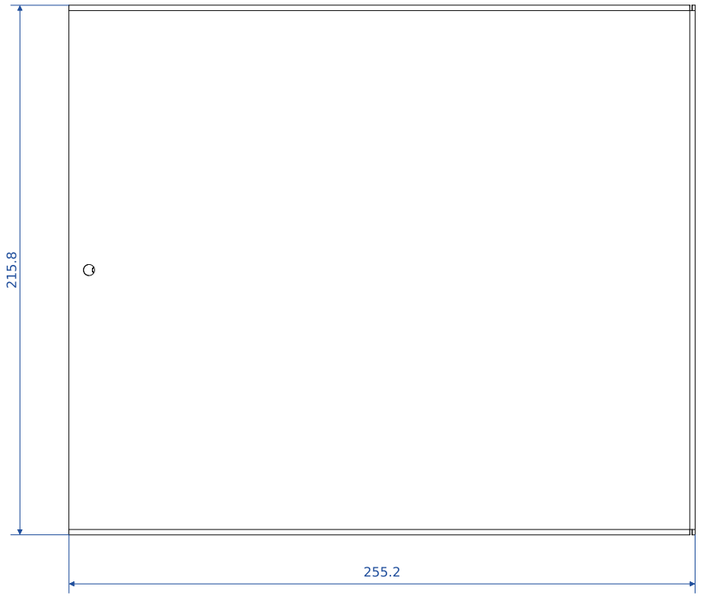
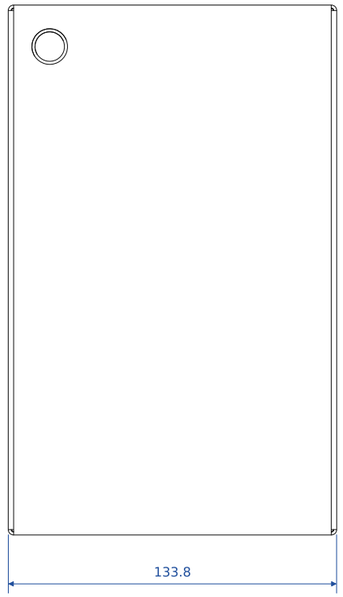
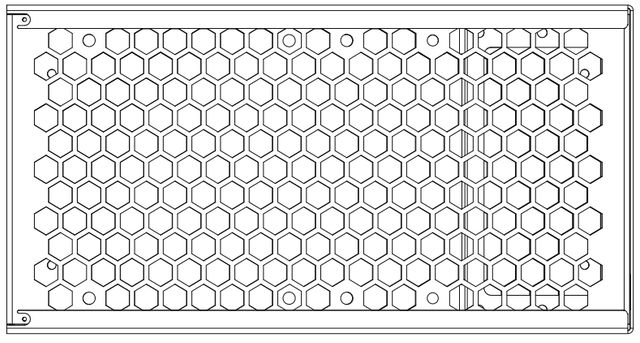
SIDE (open side; rear at left)



FRONT



TOP (from above; rear at left)



GENERAL NOTES

1. All dimensions in mm. The STEP model and the DXF flat patterns govern; do not scale this drawing.
2. Material: cold rolled low carbon steel, 1008 or equivalent, 0.048 in (1.2) thick.
3. Inside bend radius 1.0 and K = 0.516, Fabworks' figures for this sheet. The STEP models are the formed sizes to hold.
4. Tolerances unless stated: cut features +/-0.15, bend positions +/-0.4, angles +/-1 deg.
5. Deburr all edges and break sharp corners. Outside corners are R1.5 where drawn.
6. Self-clinching hardware: PennEngineering (PEM), installed by the fabricator: eleven pieces, listed below.
7. All five parts are supplied loose. The customer rivets the tray and bulkhead to the core (sheet 2) and taps the four M2 holes.
8. Finish: none ordered. If powder coated, mask threads, studs and standoff bores.

PARTS (all 0.048 in / 1.2 mm cold rolled steel)

no.	part	qty	mass	files
RMK1-01	Core: bottom, rear and top	1	711 g	core.step, core.dxf
RMK1-02	Mainboard tray	1	291 g	tray.step, tray.dxf
RMK1-03	Front bulkhead	1	244 g	bulkhead.step, bulkhead.dxf
RMK1-04	Power supply shelf	1	71 g	shelf.step, shelf.dxf
RMK1-05	Shell: front and sides	1	1371 g	shell.step, shell.dxf

HARDWARE BY THE FABRICATOR (Fabworks' ZI parts)

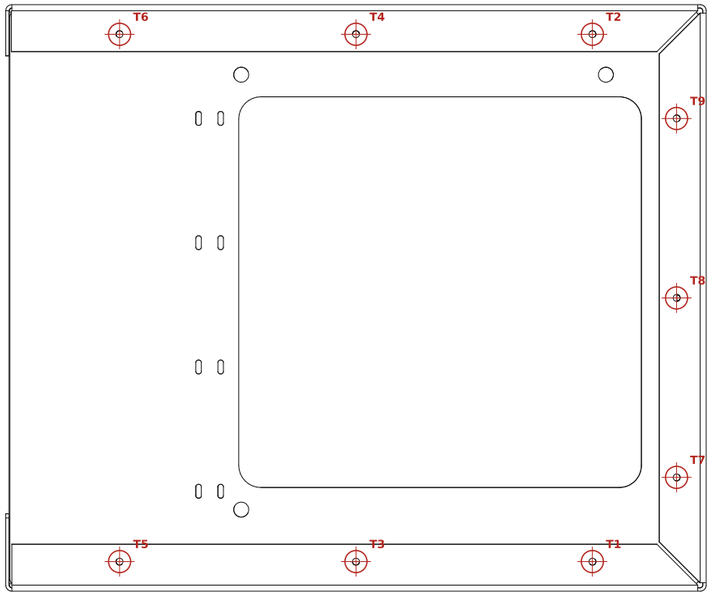
item		qty	where
PEM BSO-632-12	blind standoff, 6-32, 9.53 long	4	tray
PEM FH-632-6	flush-head stud, 6-32, 9.53 long	4	core, bottom panel
PEM S-632-1	self-clinching nut, 6-32	3	shelf 2, core 1

SCOPE

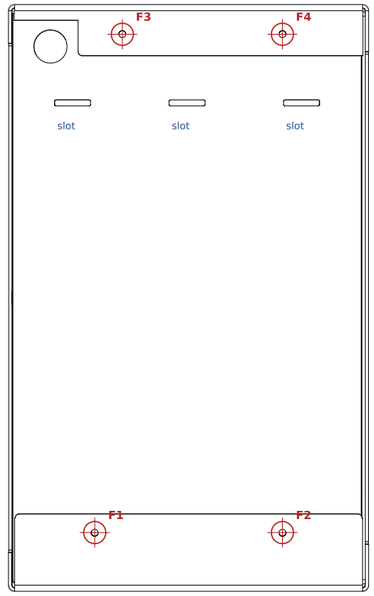
1. Five sheet metal parts, cut, bent and with their PEM hardware pressed in, all supplied loose.
2. The customer rivets the core, tray and bulkhead together (sheet 2: 13 rivets MS20426AD3-4) and taps four M2 holes.
3. Fans, mainboard, power supply, mains inlet, switch, screws and feet are fitted by the customer. The feet (sheet 8) are made separately.

TITLE	DRAWING NO.	SHEET / REV
Reactor Mk I - General arrangement	**RMK1-00**	1 of 8 rev A
MATERIAL	SCALE (A3)	DATE
see parts	1 : 2.5	2026-11-12

TRAY SIDE (front at left): 9 rivets, tray to the side lips of the core



FRONT: 4 rivets, bulkhead to front lips



RIVETS MS20426AD3-4 (centres; origin below)

	X	Y	Z
T1	40.0	tray side	9.5
T2	40.0	tray side	200.5
T3	125.7	tray side	9.5
T4	125.7	tray side	200.5
T5	211.3	tray side	9.5
T6	211.3	tray side	200.5
T7	9.5	tray side	40.0
T8	9.5	tray side	105.0
T9	9.5	tray side	170.0
F1	front	30.0	20.0
F2	front	98.0	20.0
F3	front	40.0	200.5
F4	front	98.0	200.5

RIVETING (by the customer)

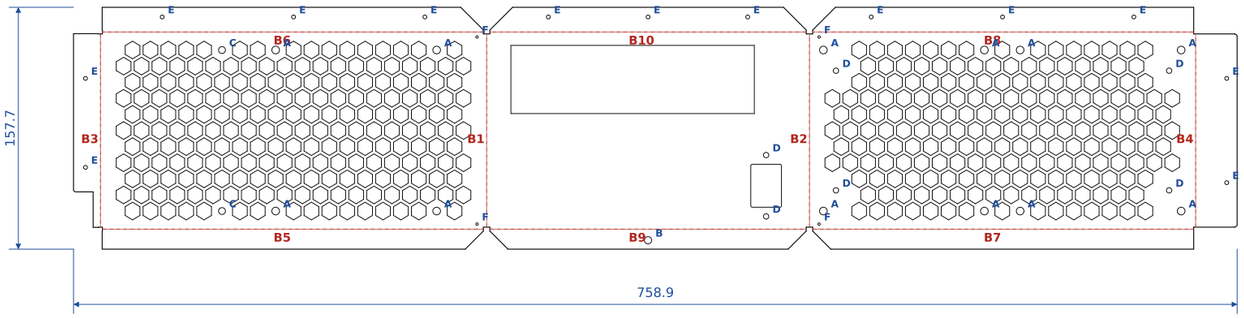
1. Datum: X from the inside face of the rear panel, Y from the inside face of the open-side lips, Z from the top face of the bottom panel.
2. Tray: its outer face against the inside of the 17 mm lips of the bottom, top and rear panels, 1.5 clear of each panel.
3. Bulkhead: its front face against the inside of the front lips of the top and bottom panels, 1.5 clear of each panel, its edge 0.3 from the tray.
4. Every rivet is 9.5 from the panel its lip is folded from, except F1 and F2 at Z 20. Lip flats are 14.8 or more.
5. Rivet: MS20426AD3-4, 3/32 in flush head, 2117 aluminium, 1/4 in long, in the 2.5 (No. 40) holes cut in both parts.
6. Countersink each hole in the core's lip from outside, 100 deg to 4.6 across, until the head sits flush or 0.1 under. Do not countersink the tray or bulkhead.
7. Head in the lip, outside. Drive the head with a flat punch while a steel bar is held against the tail inside, until the tail is 3.0 to 4.1 across and 0.8 to 1.5 tall.
8. Both sides matter: the shell passes 0.3 to 0.5 outside the lips, and the fans stand 2.8 inside the tray.
9. Rivet the tray first, then the bulkhead, starting each from its middle.
10. After riveting the open side must stay square: 128.0 +/-0.4 between the side lips, 210.0 +/-0.4 between top and bottom panels.

NOTES

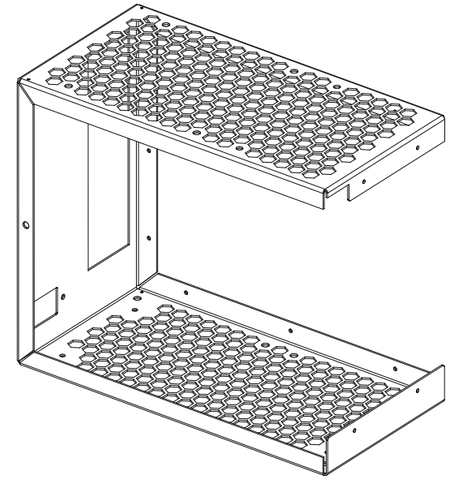
1. Outside faces of the lips are under the shell, which slides over them 0.5 clear: nothing may stand proud of them.
2. Slots marked in the front view take the shelf's three tongues, which stand 0.9 in front of the bulkhead, under the shell.

TITLE **Reactor Mk I - Riveted assembly: core, tray, bulkhead**	DRAWING NO. **RMK1-10**	SHEET / REV 2 of 8 rev A
MATERIAL RMK1-01, -02, -03	SCALE (A3) 1 : 2.22	DATE 2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in core-bends.csv)



FORMED (not to scale)



PART

Formed size	253.7 x 130.4 x 212.4
Flat size	758.9 x 157.7
Bends	10, all 90 deg, inside R1.0
Mass	711 g

ROUND HOLES

	dia	qty	for
A	5	12	fan screw, self-tapping, clearance
B	4.76	1	PEM S-632-1 nut for the shell's screw, +0.08/0
C	4.5	2	shelf screws 6-32, loose clearance
D	3.5	6	+0.08/0: PEM FH-632-6 stud, bottom panel (4); mains inlet M3, rear panel (2)
E	2.5	13	rivet, 3/32 in; countersunk by the customer
F	1.6	4	M2 x 0.4, tapped by the customer

BENDS

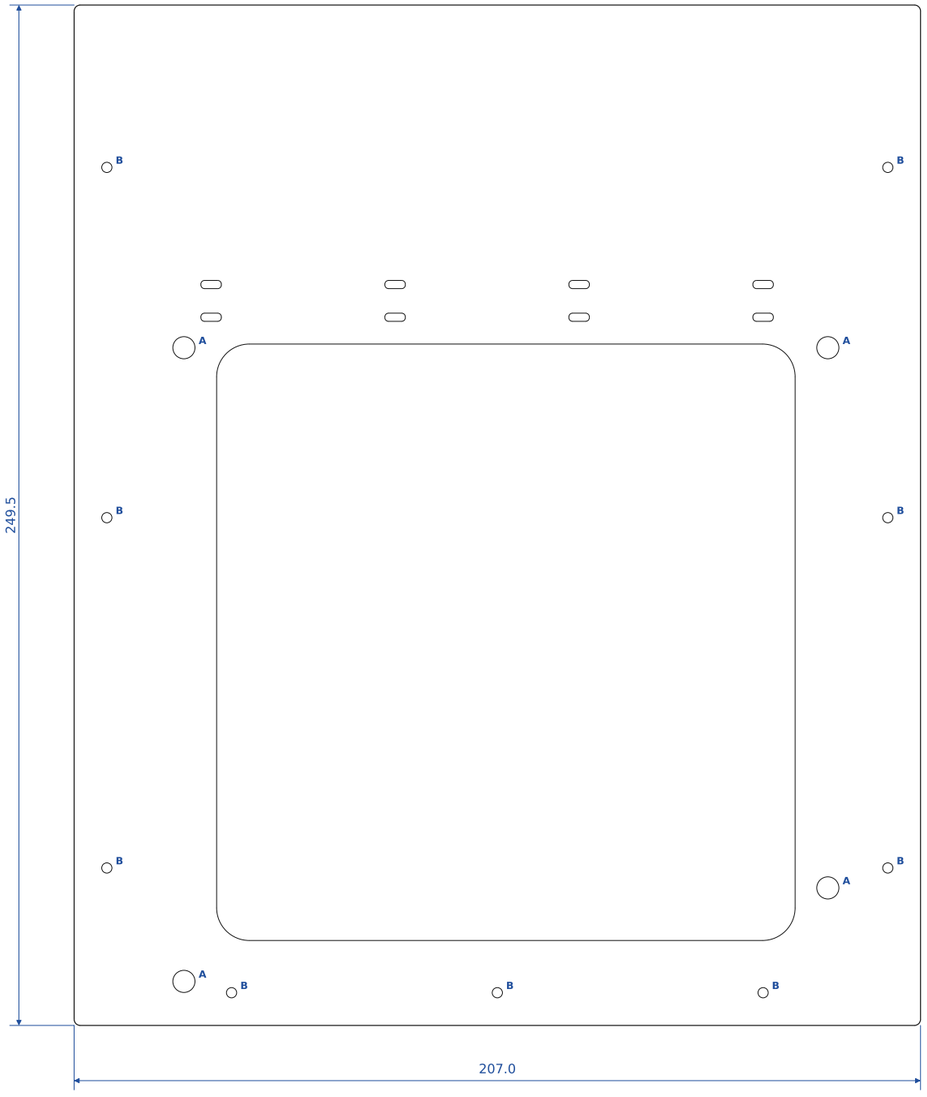
	angle	dir.	length
B1	90	up	126.0
B2	90	up	126.0
B3	90	up	126.0
B4	90	up	126.0
B5	90	up	248.5
B6	90	up	248.5
B7	90	up	248.5
B8	90	up	248.5
B9	90	up	206.3
B10	90	up	206.3

NOTES

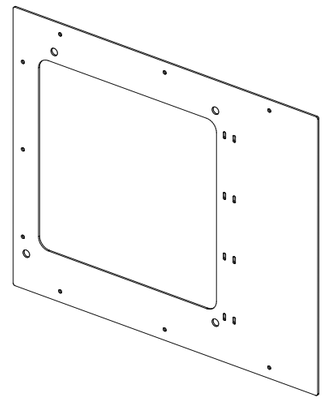
1. Hexagonal mesh: 9.6 across flats, webs 2.07; see DXF.
2. Rear panel: I/O aperture 158.75 x 44.45 +/-0.2 to ATX 2.2 fig. 4; keep paint off both faces within 2.5 of it.
3. Mains inlet cut-out 28 x 20 +0.2/0, R1, for Schurter 6100-3 mounted from outside.
4. Four 1.6 holes at the rear corners of the top and bottom panels are tapped M2 by the customer: leave them plain.
5. One PEM S-632-1 nut in the rear panel's open-side lip, flush on the outside, body inside.
6. Four PEM FH-632-6 studs in the bottom panel for the feet, heads flush in its upper (inside) face, pointing down.
7. General notes on sheet 1 apply.

TITLE **Reactor Mk I - Core: bottom, rear and top**	DRAWING NO. **RMK1-01**	SHEET / REV 3 of 8 rev A
MATERIAL steel 1008, 0.048 in	SCALE (A3) flat 1 : 4	DATE 2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in tray-bends.csv)



FORMED (not to scale)



PART

Formed size	249.5 x 1.2 x 207.0
Flat size	207.0 x 249.5
Bends	none
Mass	291 g

ROUND HOLES

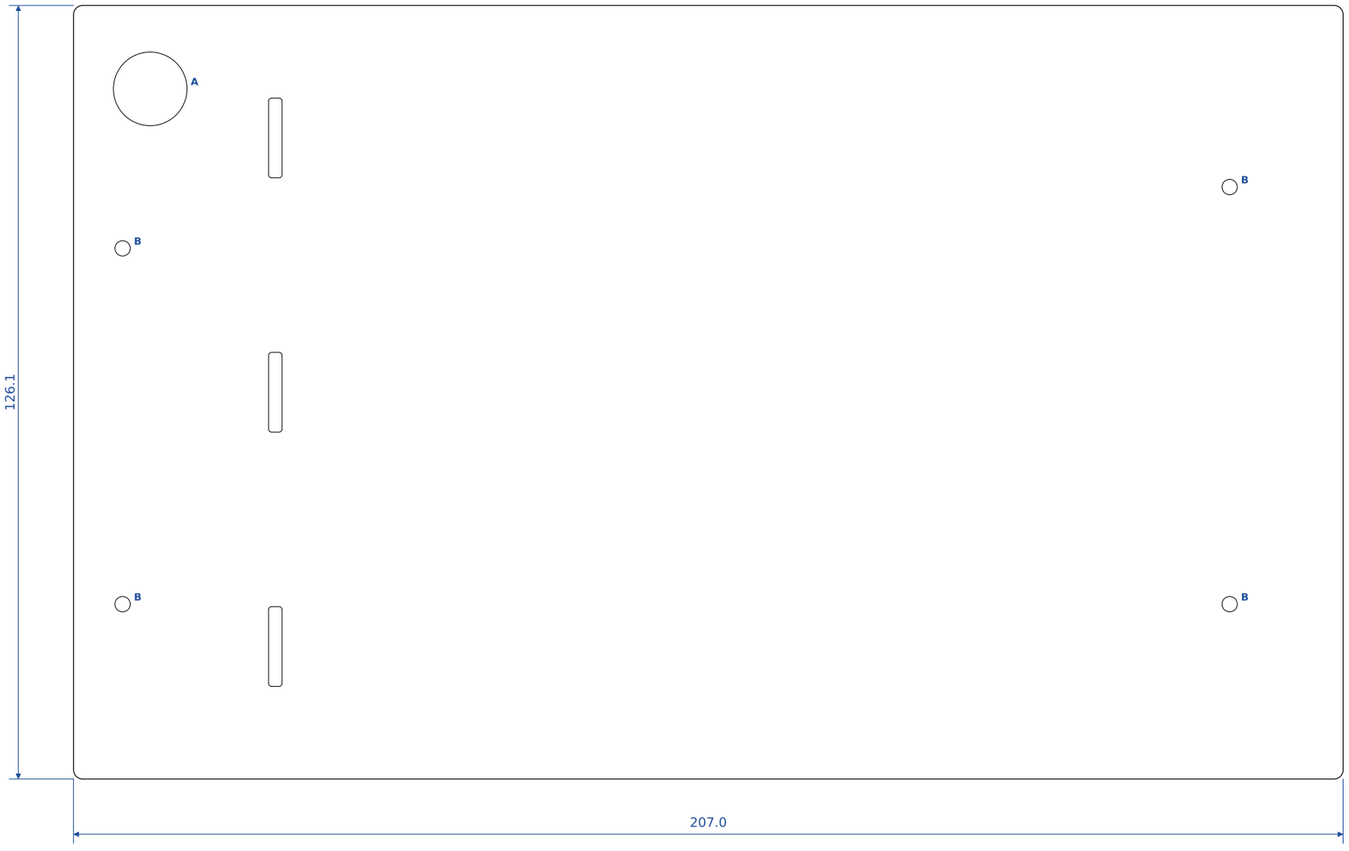
	dia	qty	for
A	5.41	4	PEM BSO-632-12 standoff, +0.08/0
B	2.5	9	rivet, 3/32 in

NOTES

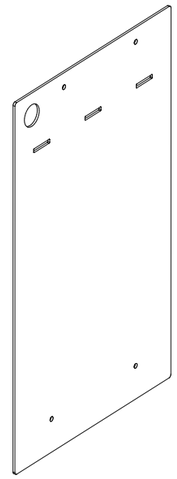
1. Flat. Four PEM BSO-632-12 standoffs pressed in from the face that goes against the core's lips, standing 9.53 off the other face.
2. The standoffs go in before the customer rivets the tray on: two heads end up under the core's lips.
3. Eight slots 2.0 x 5.0 in four pairs, for cable ties.
4. General notes on sheet 1 apply.

TITLE **Reactor Mk I - Mainboard tray**	DRAWING NO. **RMK1-02**	SHEET / REV 4 of 8 rev A
MATERIAL steel 1008, 0.048 in	SCALE (A3) flat 1 : 1.5	DATE 2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in bulkhead-bends.csv)



FORMED (not to scale)



PART

Formed size	1.2 x 126.1 x 207.0
Flat size	207.0 x 126.1
Bends	none
Mass	244 g

ROUND HOLES

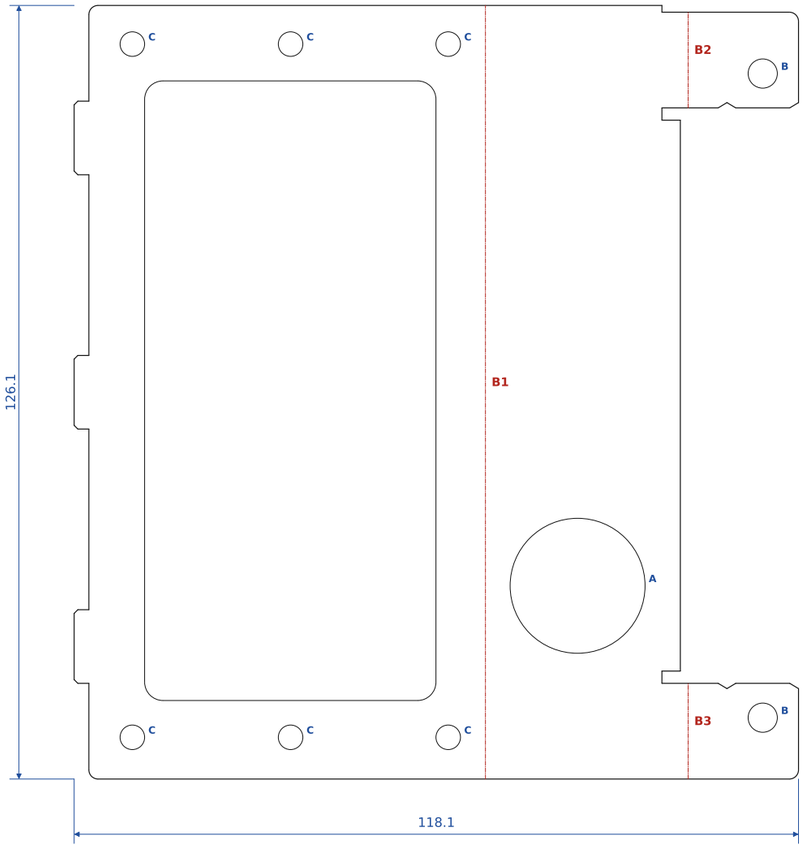
	dia	qty	for
A	12	1	power switch, +0.2/0
B	2.5	4	rivet, 3/32 in

NOTES

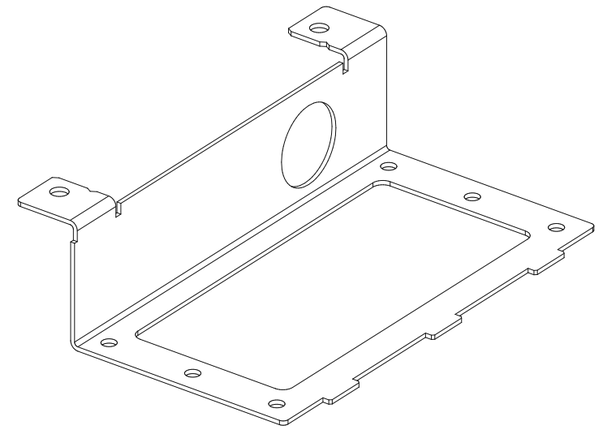
1. Flat, no hardware.
2. Three slots 13.0 x 2.2, corners R0.5, for the shelf's tongues: keep their lower edges clean and square.
3. General notes on sheet 1 apply.

TITLE **Reactor Mk I - Front bulkhead**	DRAWING NO. **RMK1-03**	SHEET / REV 5 of 8 rev A
MATERIAL steel 1008, 0.048 in	SCALE (A3) flat 1 : 1	DATE 2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in shelf-bends.csv)



FORMED (not to scale)



PART

Formed size	85.7 x 126.1 x 34.9
Flat size	118.1 x 126.1
Bends	3, all 90 deg, inside R1.0
Mass	71 g

ROUND HOLES

	dia	qty	for
A	22	1	cable grommet
B	4.76	2	PEM S-632-1 nut, +0.08/0
C	4	6	6-32 clearance, supply: SFX pattern both ways round

BENDS

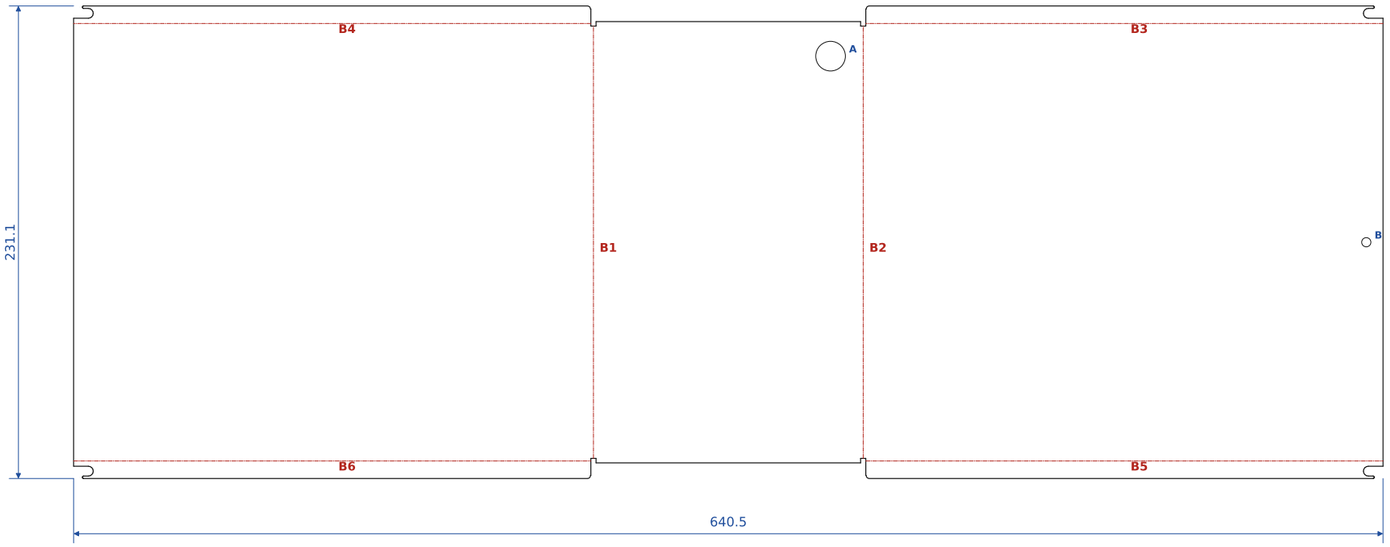
	angle	dir.	length
B1	90	up	126.1
B2	90	down	15.6
B3	90	down	15.6

NOTES

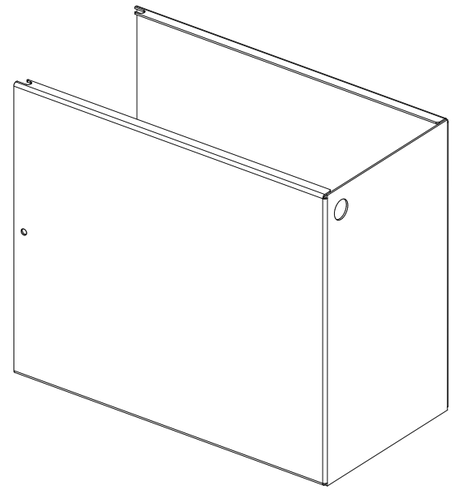
1. Two PEM S-632-1 nuts in the tabs, bodies under the tabs, flush on top. The tabs fold toward the rear, away from the supply.
2. Three tongues 12.0 wide on the front edge, standing 2.4 beyond it, go through the bulkhead's slots: no burr on them.
3. Supplied loose: its tongues sit in the bulkhead's slots and two screws through the top panel hold it.
4. General notes on sheet 1 apply.

TITLE	DRAWING NO.	SHEET / REV
Reactor Mk I - Power supply shelf	**RMK1-04**	6 of 8 rev A
MATERIAL	SCALE (A3)	DATE
steel 1008, 0.048 in	flat 1 : 1	2026-11-12

FLAT PATTERN (bend lines dash-dot, numbered as in shell-bends.csv)



FORMED (not to scale)



PART

Formed size	255.2 x 133.8 x 215.8
Flat size	640.5 x 231.1
Bends	6, all 90 deg, inside R1.0
Mass	1371 g

ROUND HOLES

	dia	qty	for
A	14.5	1	power switch head, 14
B	4.5	1	shell screw 6-32, loose clearance

BENDS

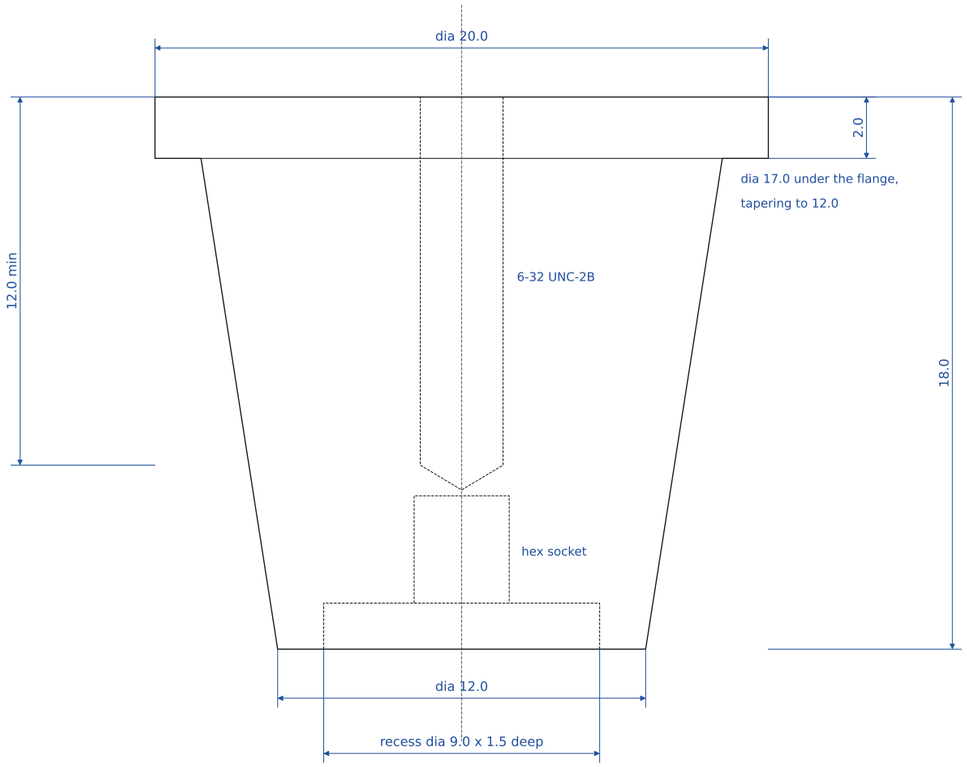
	angle	dir.	length
B1	90	up	211.4
B2	90	up	211.4
B3	90	up	253.0
B4	90	up	253.0
B5	90	up	253.0
B6	90	up	253.0

NOTES

1. Plain outside faces: no marks, grain or tooling witness on the front and sides.
2. Slots in the four return flanges run onto M2 screw heads on the core; check the slide fit before finishing.
3. General notes on sheet 1 apply.

TITLE **Reactor Mk I - Shell: front and sides**	DRAWING NO. **RMK1-05**	SHEET / REV 7 of 8 rev A
MATERIAL steel 1008, 0.048 in	SCALE (A3) flat 1 : 3	DATE 2026-11-12

SIDE (flange up, against the bottom panel; hole and recess shown hidden)



PART

Quantity	4
Model	foot.step
Mass	about 25 g
Thread	6-32 UNC-2B, 2.7 drill, 12 deep

NOTES

1. Printed in 316L stainless steel by laser powder bed fusion or binder jetting.
2. Tap 6-32 UNC-2B after printing, 10 of full thread or more. foot.step has the 2.7 tap drill hole, not the thread; drill it to size first if it prints small.
3. The flange's top face sits against the case: flat within 0.1, no support marks.
4. Hexagon socket in the recess's floor for a 3 mm key: 3.1 across flats as printed, 3.5 deep. It stops 1 short of the tapped hole.
5. Recess underneath for a stick-on bumper, 3M Bumpon SJ5076 (8.0 dia x 2.8), which stands 1.3 below the rim.
6. Each foot turns onto a PEM FH-632-6 stud standing 9.5 out of the bottom panel, tightened with the 3 mm key before the bumper goes on.
7. Finish to match the case.

TITLE	DRAWING NO.	SHEET / REV
Reactor Mk I - Foot	**RMK1-06**	8 of 8 rev A
MATERIAL	SCALE (A3)	DATE
316L stainless steel, printed	5 : 1	2026-11-12